James R. Rose, PhD

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Spatial transcriptomics · single-cell genomics · AI-driven bioinformatics

Computational biologist working at the intersection of genomics, immunology, and mRNA therapeutics. I build methods and pipelines for spatial transcriptomics, single-cell genomics, and AI-driven bioinformatics, and I run the computational function for a research group, from infrastructure to the analysis tooling lab scientists use themselves.

EXPERIENCE

Assistant Scientist

Sep 2024 – Present

Santangelo Lab, Department of Biomedical Engineering, Emory University — Atlanta, GA

Sole computational scientist for a ~25-person molecular engineering lab developing mRNA and nucleic-acid therapeutics. I own the spatial transcriptomics platform end to end, build the lab's analysis software and infrastructure, and report to program sponsors including ARPA-H, DARPA, and DTRA.

- Developed **CRISP**, an open-source R and Python package that quantifies cell segmentation quality in image-based spatial transcriptomics without ground-truth annotation, by detecting cells that co-express mutually exclusive marker genes.
- Built a multi-tissue spatial atlas of lipid nanoparticle delivery comparing injection routes across **2.8 million cells, 11 organs, and 104 tissue-microarray punches**. Route of delivery triggers distinct immune responses and remodels splenic myeloid populations.
- Built the lab's Xenium toolchain, from raw 10x output to annotated cells: a Nextflow pipeline for alternative segmentation following nf-core conventions, Seurat workflows for QC through annotation, and a Python tool that splits multi-punch tissue microarrays into per-sample datasets.
- Lead guide design for a CRISPR-activation therapeutic program. I scored and ranked a 552-guide activity screen, then checked the candidates for cross-species conservation and against gnomAD and *All of Us* population variants.
- Created **Stringency**, a safety system for AI-driven data analysis: it validates the analysis plan against best practice before any code runs, keeps a tamper-proof provenance log, and pauses at key decisions until a person signs off.
- Run the lab's AI tooling. I procured and administer the team's agent-assistant deployment, and wrote the skills that let bench scientists run validated pipelines on their own data and look up lab protocols without waiting on me.
- Administer the lab's compute: two analysis workstations with 134 cores and 180+ TB of storage, shared network mounts, and automated backups.

PhD Graduate Researcher

Aug 2019 – Aug 2024

Boss Lab, Genetics & Molecular Biology Program, Emory University — Atlanta, GA

Thesis work on how transcriptional and epigenetic state encodes immune cell identity, and how that regulation breaks in autoimmunity.

- Integrated RNA-seq and ATAC-seq across human memory T cell subsets to identify *poised accessible regions*, chromatin whose accessibility before and after activation separates the subsets and predicts how they respond. Published in *Communications Biology*.
- Showed that lupus disease activity leaves a signature in both the transcriptome and the epigenome of otherwise resting naïve B cells, in a collaboration with Emory rheumatology. Published in the *Journal of Clinical Investigation*.
- Compiled an SLE single-cell B cell reference atlas from public datasets, then queried in-house samples against it and classified disease endotypes with machine learning.

Senior Associate Scientist, Biotherapeutics Development

Aug 2016 – Jun 2019

Janssen R&D, Johnson & Johnson — Malvern, PA

- Led a team of development researchers on monoclonal antibody manufacturing process development, new technology implementation, and regulatory submissions for a therapeutic product worth $6B in annual sales.
- Implemented statistical process control across international GMP manufacturing facilities.
- Authored sections of the BLA and other regulatory filings.

Engineer I, Cell Culture Technical Development

Jun 2013 – Aug 2016

Biogen — Durham, NC

- Developed mammalian cell culture processes for monoclonal antibody production at large-scale manufacturing.
- Pioneered the group's use of stochastic (Monte Carlo) simulation modeling for manufacturing facility-fit analysis.

SOFTWARE

CRISP

R · Python

Segmentation quality assessment for spatial transcriptomics via negative-coexpression purity metrics. Supports Xenium, Seurat, and SpatialExperiment objects.

Stringency

Python

A safety harness for running AI agent data analysis pipelines. Validates the analysis plan before any code runs, keeps a tamper-proof provenance log, and pauses at key decisions until a person signs off.

Xenium analysis toolkit

Nextflow · R · Python

Three pipelines covering raw Xenium output to annotated cells: alternative segmentation (`Xen_Segmentation_NextFlow`), tissue microarray splitting (`Xen_TMA_pipeline`), and Seurat-based QC through annotation (`Xen_Seurat_Pipeline`).

SELECTED PUBLICATIONS

- **Rose, J.R.**, Rose, E.S., Assumpção, J.A.F., Pathak, H., Peck, H.E., Sasser, L.E., Patel, C.J., Vanover, D. & Santangelo, P.J. (2026) CRISP enables comparisons of image-based spatial transcriptomic segmentation quality across ten organs. *bioRxiv*2026.04.16.718947. PREPRINT
- **Rose, J.R.**, Ishii, Y., Khosroshahi, A., Hicks, S.L., Jenks, S.A., Scharer, C.D., Sanz, I. & Boss, J.M. (2026) Identification of a systemic lupus erythematosus disease signature in resting naive B cells. *Journal of Clinical Investigation*136, e199528.
- Pramod Kumar, E.K., Jung, Y., Joo, J.Y., Rose, E.S., **Rose, J.R.**, et al. & Santangelo, P.J. (2026) An engineered thio-amino polymer achieves potent, safe and broad cell type delivery in the lung mapped by spatial transcriptomics. *Advanced Materials* UNDER REVIEW
- **Rose, J.R.**, Akdogan-Ozdilek, B., Rahmberg, A.R., Powell, M.D., Hicks, S.L., Scharer, C.D. & Boss, J.M. (2023) Distinct transcriptomic and epigenomic modalities underpin human memory T cell subsets and their activation potential. *Communications Biology*6, 363.
- Patterson, D.G., Kania, A.K., Price, M.J., **Rose, J.R.**, Scharer, C.D. & Boss, J.M. (2021) An IRF4–MYC–mTORC1 integrated pathway controls cell growth and the proliferative capacity of activated B cells during B cell differentiation in vivo. *The Journal of Immunology*207(7), 1798–1811.

SKILLS

Languages & tools	R (tidyverse) · Python · Bash · Nextflow/Groovy · SQL · Git · Docker · Quarto / R Markdown · Jupyter · Shiny
Spatial & imaging	10x Xenium · Cell segmentation (Baysor · Proseg · Segger) · Spatial statistics · Tissue microarrays · Immunofluorescence integration · QuPath · ImageJ · OpenCV

Genomics	Single-cell RNA-seq · Bulk RNA-seq · ATAC-seq · CITE-seq · Seurat / Bioconductor / scverse · Differential expression (DESeq2) · Data integration · Gene regulatory networks · CRISPR guide design
Statistics & ML	Statistical modeling and inference · Machine learning · Experimental design · Monte Carlo simulation · Benchmarking · Data visualization (ggplot2 · seaborn)
AI for bioinformatics	Agentic analysis workflows · LLM tool and skill development · Reproducibility and audit frameworks · Evaluation of agent output
Infrastructure	AWS · Linux workstation and server administration · Shared storage and backup · RStudio Server · Workflow automation
Domain	Immunology (B and T cell) · Epigenetics and gene regulation · Autoimmunity · mRNA and nucleic acid delivery · Translational and clinical data · GMP biologics development

EDUCATION

PhD, Genetics & Molecular Biology
2019 – 2024

Emory University — Atlanta, GA

Advisor: Jeremy M. Boss. Defended August 2024.

Thesis *Epigenetic gene regulation in B and T lymphocytes and its role in cellular activation, differentiation, and autoimmune disease*

MS, Biotechnology
2011 – 2013

The Pennsylvania State University — University Park, PA

Integrated masters with BS degree

Thesis *Modeling interfacial shear stress on mammalian cell culture systems*

BS, Biotechnology
2008 – 2013

The Pennsylvania State University — University Park, PA

AWARDS

Abstract Award · *American Association of Immunologists Annual Meeting* 2022, 2023

AI Innovation Fellowship · *Inspirit Innovators* 2023

Undergraduate Research Award · *Eberly College of Science, Penn State* 2013

Phi Beta Kappa · *National Honors Society* 2013

LEADERSHIP & SERVICE

President & Founding Board Member
2025–present

Friends of Mulberry Fields, Atlanta GA

Founded and incorporated a 501(c)(3) that stewards a 39-plot community garden and greenspace. Drafted the bylaws, negotiated the land management agreement with the land conservancy, secured the organization's insurance and tax-exempt status, and chair the governance committee.